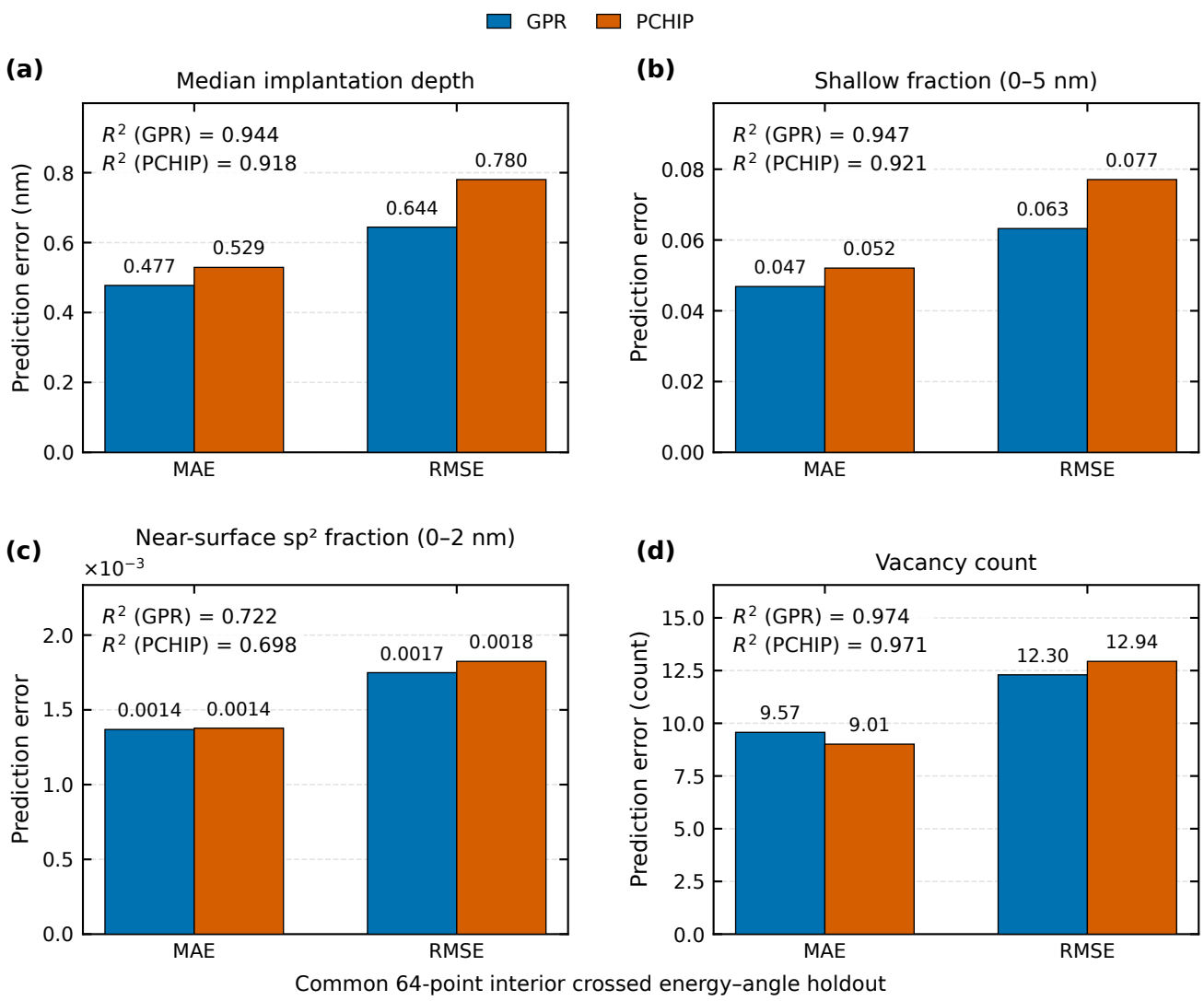
GPR
PCHIP
(a)
Median implantation depth
R^2 (GPR) = 0.944
R^2 (PCHIP) = 0.918
Prediction error (nm)
0.477
0.529
0.644
0.780
MAE
RMSE
(b)
Shallow fraction (0–5 nm)
R^2 (GPR) = 0.947
R^2 (PCHIP) = 0.921
Prediction error
0.047
0.052
0.063
0.077
MAE
RMSE
(c)
Near-surface sp² fraction (0–2 nm)
×10⁻³
R^2 (GPR) = 0.722
R^2 (PCHIP) = 0.698
Prediction error
0.0014
0.0014
0.0017
0.0018
MAE
RMSE
(d)
Vacancy count
R^2 (GPR) = 0.974
R^2 (PCHIP) = 0.971
Prediction error (count)
9.57
9.01
12.30
12.94
MAE
RMSE
Common 64-point interior crossed energy–angle holdout